

Liquid drops on vertical and inclined surfaces

II. A method for approximating drop shapes

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Abstract

In this article, a new method is proposed to approximate the shapes of liquid drops on vertical and inclined surfaces. Based on observations from Part I, the profile of a drop at a given azimuthal angle is approximated by two circles sharing a common tangent at the maximum height. The drop volume is obtained by integrating all profiles over the circumference of the base. The volume is thus described as a function of the contact angles and the three-phase contact line. The new method accurately predicts the volumes of drops tested in Part I and independent measurements from the literature. Simplifying the drop shape to a spherical cap can lead to a 75% error in drop-volume prediction. The proposed method is used to study the effect of drop parameters on volume prediction. The two-circle geometry can also be used to measure contact angles from profile images.

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1. Introduction

In many applications, it is required to calculate the total volume or area of liquid drops on solid surfaces. One of the examples of this problem is estimating the amount of condensed water accumulating on a heat exchanger. Another example is finding the quantity of pesticides retained on plant leaves. The adherence of sprayed paint to surfaces represents another manifestation of the problem. In order to accurately determine the volume of a drop, it is essential to understand its shape. A liquid drop on a horizontal plane generally takes the shape of a spherical cap. In this case, the three-phase contact line at the base of the drop is circular, and the contact angle is constant along the circumference. If the plane is inclined, however, the vapor–liquid interface of the drop cannot be described by a simple shape. The change in drop shape is manifested in an elongation of the drop contour and a modification to the contact angles within the drop.

The shape of the vapor–liquid boundary of a drop can be determined from the Young–Laplace (capillarity) equation, which relates the pressure difference across the interface to

the surface tension and curvature. For a drop on an inclined plane, the pressure across the interface can be found from hydrostatic forces, and the resulting differential equation can be solved for the location of any point on the liquid surface. Larkin [1] used the finite-difference method to solve the capillarity equation using the contact angle as a boundary condition. The contact angle was assumed to equal that of a horizontal surface at a single point where the normal to the contour is perpendicular to gravity. The calculations resulted in the horizontal contact angle, θ_H , having an intermediate value between a minimum and a maximum. Brown et al. [2] presented a finite-element solution to the Young–Laplace equation to find shapes of drops on planes with various inclinations. The contact line at the base of the drop was assumed to be fixed to the solid surface and to have a circular shape. The numerical computations showed the horizontal contact angle to be almost an average of the minimum and maximum angles of the drop. This work also predicted the variation of the contact angle along the base of the drop. The contact angle hysteresis for a drop increased with the surface inclination to as much as 145° at an inclination angle of 60° .

Another numerical study by Milinazzo and Shinbrot [3] minimized the surface and gravitational energy for a drop of a fixed volume on a vertical wall. They presented their computations in terms of the Bond number, Bo , which is a

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ratio of gravitational to surface-tension forces,

$$Bo = \frac{\rho g D^2 \sin \alpha}{\gamma}, \quad (1)$$

where ρ is the liquid density, g is the acceleration of gravity, D is the drop diameter, α is the surface inclination angle, and γ is the surface tension of the liquid–vapor interface. Milinazzo and Shinbrot assumed that the drop contour had a fixed area, while the Bond number increased to a maximum. However, a drop on a vertical plane with fixed area has a constant Bond number. Therefore, the increase in Bond number in their work may be reinterpreted as an increase in the inclination angle of the plane from 0° to 90° for the same drop. Milinazzo and Shinbrot found the contact angle hysteresis to increase with the Bond number until either the advancing contact angle, θ_A , reaches 180° or the receding contact angle, θ_R , reaches 0° . The increase in θ_A and decrease in θ_R from the horizontal contact angle were almost linear functions of the Bond number. With the new interpretation of the data, the contact angle results are similar in nature to those of Brown et al. [2].

Rotenberg et al. [4] solved the drop-shape problem by minimizing the energy of the drop using a finite-element method. They prescribed the variation of the contact angle along the contour as a boundary condition. In defining contact angles, they assumed the drop was moving slowly and imposed relations between dynamic contact angles and the normal velocity of the contact line. Rotenberg and co-workers also presented experimental observations of contours and median profiles of drops on inclined planes and concluded that a circular contour is a reasonable assumption for static drops.

Merte and Yamali [5] considered two-dimensional drops on vertical surfaces. They minimized the total energy of drops for given volumes, using variational calculus. The profile at the symmetry plane was assumed to have infinite width, and the base length was taken to represent drop diameter. The advancing contact angle and drop diameter were input, while the receding angle and profile were output. The two-dimensional analysis neglects the variation of contact angles along the drop contour, which affects surface-tension forces. Moreover, for noncircular contours, the base length at the symmetry plane is not equal to the diameter of the drop. Merte and Son [6] extended the work to include advancing angles of up to 180° .

Dimitrakopoulos and Higdon [7] considered drops on the verge of sliding on inclined planes. They minimized the contact-angle hysteresis for a given advancing angle and Bond number, with the condition that pressures from surface tension and gravity are equal at the liquid surface. Using this approach, no conditions were imposed on the shape of the drop contour. However, the resulting drop contours showed elongation in the direction perpendicular to gravity, which is contrary to experimental observations. Therefore, Dimitrakopoulos and Higdon imposed an additional condition to prevent side-expansion of the contact line. The results

showed the contact-angle variation along the drop base to almost follow a step function between θ_A and θ_R .

The investigations discussed above either imposed or obtained questionable contact angles. The horizontal contact angle, θ_H , was found in [1–3] to have an intermediate value between θ_A and θ_R . This result is in contrast with the experimental observations of MacDougall and Ockrent [8], who found the advancing angle at detachment conditions to be equal to the horizontal contact angle for various surface inclinations, except for some cases where it was only slightly higher than θ_H . Kwok and Neumann [9] agreed with this finding. In addition to the problematic θ_A results, drop shape studies have disagreed on how the contact angles vary along the drop base, as in [2–4] and [7]. The contact-angle discrepancies, along with different drop-base contours as reported by Extrand and Kumagai [10], lead to the conclusion that available models for the shapes of drops on inclined surfaces are not sufficiently reliable. Consequently, the drop volume cannot be accurately determined. Although there have been some efforts to determine maximum volumes of drops on inclined surfaces, no reliable function relating the volume of a general drop to its shape parameters is found in the literature.

The objective of this work is to develop a method to describe the three-dimensional shape of a general drop on a plane surface, given its diameter and contact angles, and use it to obtain drop volume and study shape parameters. The method is based on empirical approximations of the drop profiles from experiments reported in a companion paper [11]. In the next section, a special case is considered to relate the approximate profile to the physics of the problem. The method is then explained and discussed afterwards.

2. Drop profile on a vertical plane surface

Consider a drop on a vertical plane, the profile of which is shown in Fig. 1. Let the contact angles of this section be denoted by θ_1 and θ_2 , as shown. The Young–Laplace equation

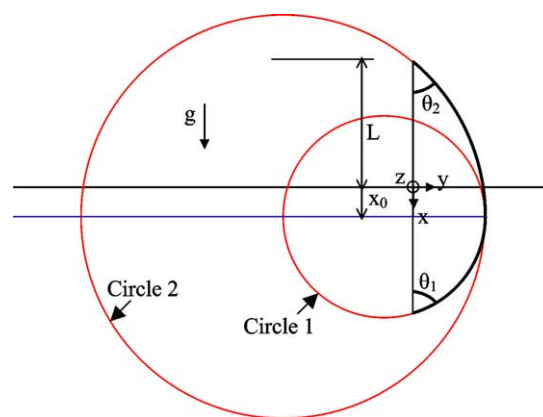


Fig. 1. The midsection of a liquid drop on a vertical plane with contact angles θ_1 and θ_2 . The z -axis points out of the page. The length at the drop base is $2L$.

may be written for any point on the liquid–vapor interface,

$$\Delta P = \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right), \quad (2)$$

where ΔP is the pressure difference across the interface, γ is the liquid surface tension, and R_1 and R_2 are the principal radii of curvature. The pressure in Eq. (2) may be obtained from the hydrostatic pressure,

$$\Delta P = \Delta \rho g x + \Delta P_0, \quad (3)$$

where $\Delta \rho$ is the density difference across the interface, g is the acceleration of gravity, x is the vertical coordinate of the point, and ΔP_0 is the pressure difference at the origin ($\Delta P_0 = \Delta \rho g L$). Equating the pressures in Eqs. (2) and (3), we can get an expression for the mean curvature at any point on the surface of the drop,

$$\frac{1}{R_1} + \frac{1}{R_2} = c_1 x + c_2, \quad (4)$$

where $c_1 = \Delta \rho g / \gamma$ and $c_2 = \Delta P_0 / \gamma$.

The surface of the drop can be expressed as a function $Y = Y(x, z)$. Let the subscripts x and z denote partial derivatives with respect to x and z , respectively. For the special two-dimensional case where the drop shape is similar to a “speed bump,” the mean curvature can be expressed as

$$\frac{1}{R_1} + \frac{1}{R_2} = \frac{Y_{xx}}{(1 + Y_x^2)^{3/2}}. \quad (5)$$

Combining the previous two equations, a differential equation for the profile in Fig. 1 is obtained,

$$\frac{Y_{xx}}{(1 + Y_x^2)^{3/2}} = c_1 x + c_2, \quad (6)$$

with the following boundary conditions:

$$\text{at } x = L, \quad Y_x = -\tan \theta_1; \quad (7a)$$

$$\text{and at } x = -L, \quad Y_x = \tan \theta_2. \quad (7b)$$

The negative sign in condition (7a) is due to θ_1 being measured through the liquid. The conditions also apply if the contact angles are greater than 90° . Using the substitution $\omega = Y_x$ and integrating, Eq. (6) becomes

$$\int \frac{1}{(1 + \omega^2)^{3/2}} d\omega = \int (c_1 x + c_2) dx. \quad (8)$$

Performing the integration and substituting back, the equation becomes first order:

$$\frac{Y_x}{\sqrt{1 + Y_x^2}} - c_4 x = \frac{c_1}{2} x^2 + c_2 x + c_3. \quad (9)$$

Rearranging, Y_x becomes

$$Y_x = -\frac{\frac{1}{2} c_1 x^2 + c_5 x + c_3}{\sqrt{1 - (\frac{1}{2} c_1 x^2 + c_5 x + c_3)^2}}, \quad (10)$$

where $c_5 = c_2 + c_4$.

Applying the slope boundary condition (7a) and using the identity $(\tan^2 \theta + 1) = \sec^2 \theta$, results in

$$\frac{1}{2} c_1 L^2 + c_5 L + c_3 = \sin \theta_1. \quad (11)$$

Similarly, the boundary condition (7b) leads to

$$\frac{1}{2} c_1 L^2 - c_5 L + c_3 = -\sin \theta_2. \quad (12)$$

Solving the two equations, we get the constants

$$c_3 = \frac{1}{2} (\sin \theta_1 - \sin \theta_2 - \Delta \rho g / \gamma L^2), \quad (13a)$$

$$c_5 = (\sin \theta_1 + \sin \theta_2) / 2L, \quad \text{and} \quad (13b)$$

$$c_1 = \Delta \rho g / \gamma, \quad \text{as defined earlier.} \quad (13c)$$

Setting Y_x in Eq. (10) equal to zero, we can obtain the location x_0 where the drop “height” (in the y -direction) reaches a maximum,

$$x_0 = \frac{-c_5 + \sqrt{c_5^2 - 2c_1 c_3}}{c_1}, \quad (14)$$

where the constants are as defined in (13).

Since Eq. (10) has no closed-form solution, Y_x of the drop profile is compared to Y_x of two circles having a common tangent at the maximum value of the y -coordinate. Let Circle 1 be centered at (x_0, y_{01}) and described by the equation

$$(x - x_0)^2 + (y - y_{01})^2 = r_1^2. \quad (15)$$

The derivative at a point on Circle 1 becomes

$$Y_{x1} = -\frac{x - x_0}{\sqrt{r_1^2 - (x - x_0)^2}}. \quad (16)$$

Similarly, the derivative for Circle 2, which is centered at (x_0, y_{02}) with a radius r_2 , is

$$Y_{x2} = -\frac{x - x_0}{\sqrt{r_2^2 - (x - x_0)^2}}, \quad (17)$$

where $r_1 = (L - x_0) / \sin \theta_1$ and $r_2 = (L + x_0) / \sin \theta_2$. Fig. 2 shows Y_x plotted against x for the drop profile (Eq. (10)), compared to Y_{x1} and Y_{x2} for the circles. Circle 1 closely approximates the profile of the right portion ($x > x_0$) and deviates for $x < x_0$. The left portion of the profile ($x < x_0$) is closely approximated by Circle 2. Thus, the two circles, taken together, provide a good approximation for this profile.

Experiments described in [11] and [12] indicate that the two-circle approximation works well for more general sections. If the section in Fig. 2 is taken at other azimuthal angles or different surface inclinations, the profile can still be fit by two circles. Fig. 3 shows a typical image of a drop profile fit by two circles. Additional images of drop profiles can be found in [12]. The next section describes the two-circle method in more detail.

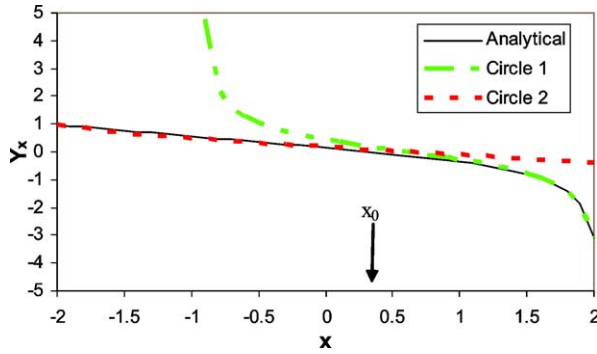


Fig. 2. Derivative of the profile of a drop on a vertical plane, compared to the corresponding derivatives of two circles approximating the profile. Circle 1 closely approximates the right portion of the profile, while Circle 2 approximates the left portion. $L = 2$, $\theta_1 = 72^\circ$, and $\theta_2 = 44^\circ$.

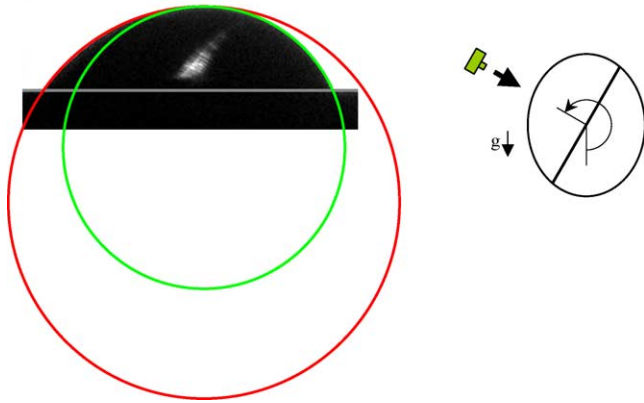


Fig. 3. Typical profile of a drop on an inclined surface fit by two circles. The image is taken from an azimuthal angle of 240° . The drop volume is 5 mm^3 and the surface is inclined 45° .

3. The profile of an arbitrary section

Consider a drop where the base of the drop is inclined with respect to gravity, as shown in Fig. 4. For this cross section, the length at the base of the drop is given as 2ζ , and the contact angles θ_1 and θ_2 are known. The drop contour can have a general shape, with D representing the equivalent diameter and 2ζ being the length at the current cross section. Since the x -component of gravity points in the positive direction, $\theta_1 \geq \theta_2$. The two circles meet at the highest point of the profile, where $x = X_0$ and y is a maximum. The circle with a contact angle θ_1 is named C1 and variables with subscript 1 refer to it. Similarly, variables with subscript 2 refer to the other circle, C2. The origin is selected so that C2 is centered at $(X_0, 0)$ and C1 is centered at (X_0, Y_0) , as shown in Fig. 4. Equating the height of the drop from C1 and C2,

$$\frac{L_1}{\sin \theta_1} - \frac{L_1}{\sin \theta_1} \cos \theta_1 = \frac{L_2}{\sin \theta_2} - \frac{L_2}{\sin \theta_2} \cos \theta_2. \quad (18)$$

Simplifying, L_1 can be expressed as

$$L_1 = L_2 L_f, \quad (19)$$

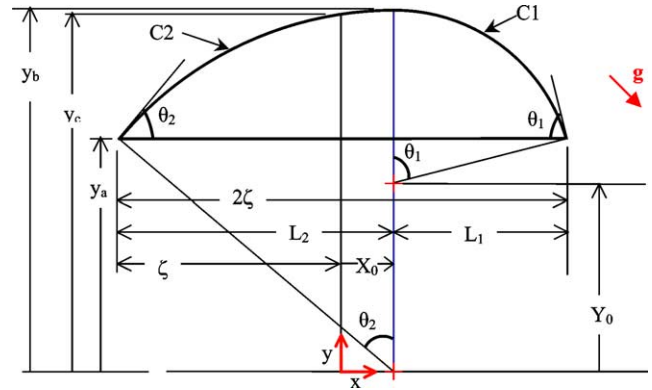


Fig. 4. The profile of a drop on an inclined plane with the nomenclature for the two-circle method.

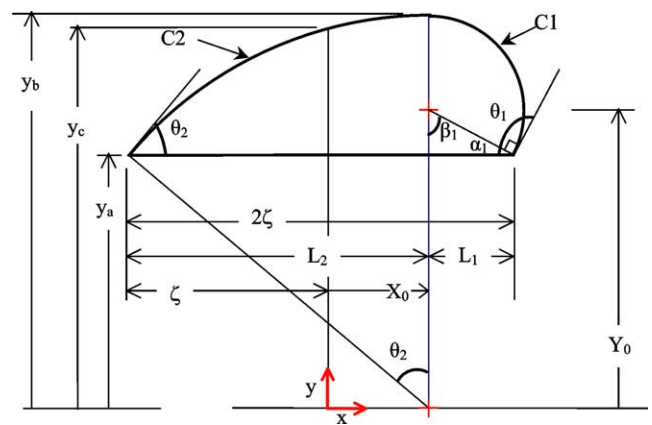


Fig. 5. The profile of a drop with one contact angle greater than 90° . The equations developed to describe the profile apply also to cases where one or both angles exceed 90° .

where

$$L_f = \frac{\sin \theta_1 (1 - \cos \theta_2)}{\sin \theta_2 (1 - \cos \theta_1)}. \quad (20)$$

Since $L_1 + L_2 = 2\zeta$, the variables L_1 and L_2 can be expressed as

$$L_1 = \frac{2\zeta L_f}{1 + L_f} \quad (21a)$$

and

$$L_2 = \frac{2\zeta}{1 + L_f}. \quad (21b)$$

The ratio L_f depends only on the contact angles. Therefore, L_1 and L_2 can be obtained from Eq. (21), if the base length and contact angles are known.

A few observations can be made at this point. The function $\sin \theta / (1 - \cos \theta)$ is a decreasing positive function for $0^\circ < \theta < 180^\circ$ and $\theta_1 \geq \theta_2$, as mentioned earlier. Therefore, $L_f \leq 1$, which leads to the conclusion that $L_1 \leq L_2$. In other words, $X_0 \geq 0$. This result also indicates that the radius of C1 is smaller or equal to that of C2, and $Y_0 \geq 0$. Another observation is regarding the case when one or both contact angles are greater than 90° . Fig. 5 shows the geometry for a case of $\theta_1 > 90^\circ$. It can be seen from the figure that

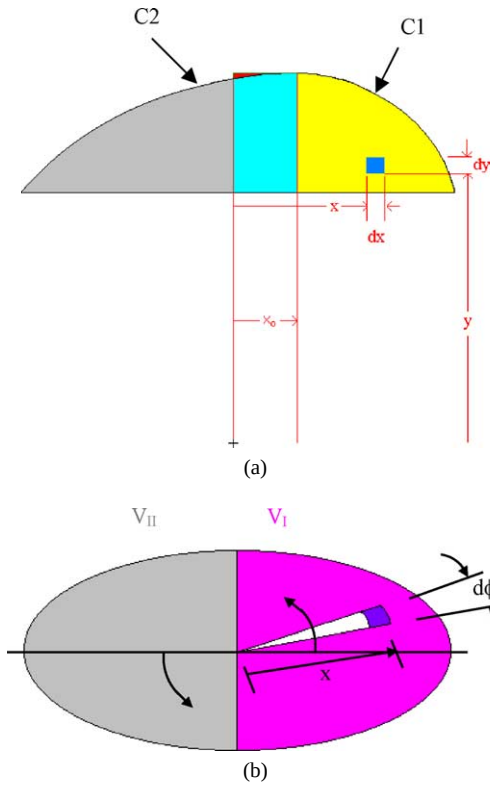


Fig. 6. Side (a) and top (b) views showing a differential element used in cylindrical coordinates to obtain the volume of a drop by integration.

$\sin \beta_1 = \sin \theta_1$ and $\cos \beta_1 = -\cos \theta_1$. Thus, the drop height based on C1 of Fig. 5 is

$$\frac{L_1}{\sin \beta_1} + \frac{L_1}{\sin \beta_1} \cos \beta_1 = \frac{L_1}{\sin \theta_1} - \frac{L_1}{\sin \theta_1} \cos \theta_1. \quad (22)$$

Therefore, the same expressions apply when one or both angles are greater than 90° .

4. The volume of a drop

The volume of a drop may be obtained by integrating the profiles over cross sections taken at all azimuthal angles. Fig. 6 shows the side and top views of a differential element used in the volume integration. The cylindrical coordinate system is adopted, with the conventional (r, θ, z) replaced by (x, ϕ, y) . The volume of the drop can be expressed by the triple integral

$$V = \int \int \int_V x dx dy d\phi. \quad (23)$$

The drop is assumed to be symmetric around the plane $\phi = 0$, since the gravity vector is parallel to that plane.

In the integration of Eq. (23), the drop is divided into two parts, V_I for $-\pi/2 \leq \phi \leq \pi/2$, and V_{II} for $\pi/2 < \phi < 3\pi/2$,

as shown in Fig. 6. Thus,

$$V_I = 2 \int_0^{\pi/2} \int_{y_a}^{y_b} \int_{x_0}^{x_1} x_1 dx_1 dy_1 d\phi + 2 \int_0^{\pi/2} \left[\int_{y_a}^{y_b} \int_0^{x_0} x_0 dx_1 dy_1 - \int_{y_c}^{y_b} \int_0^{x_2} x_2 dx_2 dy_2 \right] d\phi, \quad (24a)$$

$$V_{II} = 2 \int_{\pi}^{3\pi/2} \int_{y_a}^{y_c} \int_0^{x_2} x_2 dx_2 dy_2 d\phi, \quad (24b)$$

and

$$V = V_I + V_{II}. \quad (24c)$$

The first term of Eq. (24a) represents the part of V_I generated by C1, where the geometry of the profile changes from one section to another. In the second term, the area between the base of the drop and the line $y = y_b$ is integrated over the drop base, with $0 \leq x \leq X_0$. The third term subtracts the area between that line and C2. Equation (24b) gives an expression for the portion of C2 generating V_{II} . Notice that X_0 always lies in the V_I portion of the drop, as discussed in the previous section.

The variables y_a , y_b , and y_c used in the integration limits are the y -coordinates of the profile at, respectively, the drop base, the maximum height (at $x = X_0$), and $x = 0$. These coordinates, which are shown in Fig. 4, can be expressed as

$$y_a = \frac{L_2}{\sin \theta_2} \cos \theta_2 = Y_0 + \frac{L_1}{\sin \theta_1} \cos \theta_1, \quad (25a)$$

$$y_b = \frac{L_2}{\sin \theta_2} = Y_0 + \frac{L_1}{\sin \theta_1}, \quad (25b)$$

$$y_c = \sqrt{\frac{L_2^2}{\sin^2 \theta_2} - X_0^2}, \quad (25c)$$

where

$$X_0 = \zeta - L_1. \quad (25d)$$

Notice that the distance between the origin and the drop base, y_a , changes from one section to another. However, since the differential element is integrated around the y -axis, the location of the origin on the y -axis does not affect the volume integration.

Equation (24a) may be rewritten as

$$V_I = \int_0^{\pi/2} T_{11} d\phi + \int_0^{\pi/2} T_{121} d\phi - \int_0^{\pi/2} T_{122} d\phi, \quad (26)$$

where

$$T_{11} = \int_{y_a}^{y_b} \int_{X_0}^{x_1} 2x_1 dx_1 dy_1 = \int_{y_a}^{y_b} (x_1^2 - X_0^2) dy_1. \quad (27)$$

But the equation of circle C1 can be used to express x_1 as

$$x_1 = \sqrt{\frac{L_1^2}{\sin^2 \theta_1} - (y_1 - Y_0)^2} + X_0. \quad (28)$$

Substituting (28) into (27) and integrating, the term T_{11} becomes

$$T_{11} = \frac{L_1^3}{3 \sin^3 \theta_1} (2 - 3 \cos \theta_1 + \cos^3 \theta_1) + \frac{X_0 L_1^2}{\sin \theta_1} \left(\frac{\theta_1}{\sin \theta_1} - \cos \theta_1 \right). \quad (29)$$

The second term on the right hand side of Eq. (26) is

$$T_{121} = \int_{y_a}^{y_b} \int_0^{X_0} 2X_0 dx_1 dy_1 = X_0^2 \frac{L_2}{\sin^2 \theta_2} (1 - \cos \theta_2). \quad (30)$$

The equation for the third term is

$$T_{122} = \int_{y_c}^{y_b} \int_0^{X_0} 2x_2 dx_2 dy_2 = \int_{y_a}^{y_b} x_2^2 dy_2. \quad (31)$$

For the portion of C2 in V_I , x_2 can be expressed as

$$x_2 = X_0 - \sqrt{\frac{L_2^2}{\sin^2 \theta_2} - y_2^2}. \quad (32)$$

Substituting (32) into (31), integrating, and simplifying, the expression for T_{122} becomes

$$T_{122} = \frac{2}{3} \frac{L_2^3}{\sin^3 \theta_2} - \frac{2}{3} \frac{L_2^2}{\sin^2 \theta_2} \sqrt{\frac{L_2^2}{\sin^2 \theta_2} - X_0^2} + \frac{X_0^2 L_2}{\sin \theta_2} - \frac{X_0^2}{3} \sqrt{\frac{L_2^2}{\sin^2 \theta_2} - X_0^2} - \frac{X_0 L_2^2}{\sin^2 \theta_2} \left(\frac{\pi}{2} - \sin^{-1} \left(\sqrt{1 - \frac{X_0^2}{L_2^2/\sin^2 \theta_2}} \right) \right). \quad (33)$$

Since x is a radial coordinate, x_2 in the V_{II} portion of the drop is positive and can be expressed as

$$x_2 = \sqrt{\frac{L_2^2}{\sin^2 \theta_2} - y_2^2} - X_0. \quad (34)$$

Equation (34) can be substituted into (24b) to get

$$V_{II} = \int_{\pi}^{3\pi/2} \int_{y_a}^{y_c} \left[\left(\frac{L_2^2}{\sin^2 \theta_2} - y_2^2 \right) - 2X_0 \sqrt{\frac{L_2^2}{\sin^2 \theta_2} - y_2^2} + X_0^2 \right] dy_2 d\phi, \quad (35)$$

which can be written as

$$V_{II} = \int_{\pi}^{3\pi/2} T_{21} d\phi + \int_{\pi}^{3\pi/2} T_{22} d\phi + \int_{\pi}^{3\pi/2} T_{23} d\phi. \quad (36)$$

After integrating and simplifying, the terms in Eq. (36) become

$$T_{21} = \sqrt{\frac{L_2^2}{\sin^2 \theta_2} - X_0^2} \left(\frac{2}{3} \frac{L_2^2}{\sin^2 \theta_2} + \frac{X_0^2}{3} \right) - \frac{L_2^3}{\sin^3 \theta_2} \left(\cos \theta_2 - \frac{\cos^3 \theta_2}{3} \right), \quad (37a)$$

$$T_{22} = -X_0^2 \sqrt{\frac{L_2^2}{\sin^2 \theta_2} - X_0^2} + \frac{X_0 L_2^2}{\sin \theta_2} \cos \theta_2 - \frac{X_0 L_2^2}{\sin^2 \theta_2} \left(\sin^{-1} \left(\sqrt{1 - \frac{X_0^2}{L_2^2/\sin^2 \theta_2}} \right) - \frac{\pi}{2} + \theta_2 \right), \quad (37b)$$

and

$$T_{23} = X_0^2 \sqrt{\frac{L_2^2}{\sin^2 \theta_2} - X_0^2} - \frac{X_0^2 L_2 \cos \theta_2}{\sin \theta_2}. \quad (37c)$$

As mentioned earlier, adding V_I and V_{II} results in the total volume of the drop. The equations describing the volume of a drop apply also for the case when θ exceeds 90° .

The foregoing derivations allow describing the volume of a liquid drop on a plane surface. To summarize, the total volume of the drop is obtained by adding two volumes, V_I and V_{II} . The volume V_I is determined from Eqs. (26), (29), (30), and (33), which can be put in a shorter form as

$$V_I = \int_0^{\pi/2} \left[\frac{(y_b - Y_0)^3}{3} (2 - 3 \cos \theta_1 + \cos^3 \theta_1) + X_0 (y_b - Y_0)^2 \theta_1 - X_0 (y_a - Y_0) (y_b - Y_0) \sin \theta_1 + X_0^2 (y_b - y_a) - \frac{2}{3} y_b^3 + \frac{2}{3} y_b^2 y_c - X_0^2 y_b + \frac{X_0^2}{3} y_c + X_0 y_b^2 \left(\frac{\pi}{2} - \sin^{-1} \left(\frac{y_c}{y_b} \right) \right) \right] d\phi. \quad (38)$$

Similarly, the volume V_{II} is obtained from Eqs. (36) and (37), which can be expressed as

$$V_{II} = \int_{\pi}^{3\pi/2} \left[y_c \left(\frac{2}{3} y_b^2 + \frac{X_0^2}{3} \right) - y_b^2 y_a + \frac{y_a^3}{3} - X_0^2 y_a + X_0 y_a y_b \sin \theta_2 - X_0 y_b^2 \left(\sin^{-1} \left(\frac{y_c}{y_b} \right) - \frac{\pi}{2} + \theta_2 \right) \right] d\phi, \quad (39)$$

where y_a , y_b , y_c , and X_0 are determined from Eqs. (25), which require knowledge of L_1 , L_2 , and L_f from Eqs. (20) and (21). Notice that all the variables in Eqs. (38) and (39) can be expressed in terms of ζ , θ_1 , and θ_2 . Therefore, Eqs. (38) and (39) provide an approximation to the volume of a drop on a plane surface, which is expressed as a function of known drop parameters in the form

$$V \approx \int f(\zeta(\phi), \theta(\phi)) d\phi. \quad (40)$$

The development of this method imposed no restrictions on the contact angle function $\theta(\phi)$ or the contour shape $\zeta(\phi)$, and kept them as inputs. Thus, the method allows different models for these functions to be used and compared. As mentioned in the Introduction, earlier studies reported different shapes for each of the contact angle and contour functions. These two functions have been experimentally investigated in Part I [11]. With ζ and θ described as functions of ϕ , Eq. (40) can be integrated to get the volume of the drop.

5. Discussion

5.1. Method validation

The proposed method is based on approximating the profile of a cross section by parts of two circles sharing a tangent. The analysis presented earlier provided a physical basis for this approximation, although only a special case was considered analytically. Images of profiles at different azimuthal angles and surface inclinations, shown and discussed in [12], verify that two circles can reasonably fit a profile in the manner described. Notice that for a section perpendicular to the symmetry plane, the two circles coincide, forming one circle. In this case, the contact angles of the section, θ_1 and θ_2 , become equal. The difference between θ_1 and θ_2 reaches a maximum at the plane $\phi = 0$, where $\theta_1 = \theta_{\max}$ and $\theta_2 = \theta_{\min}$.

In order to validate the proposed method, it is used to calculate volumes of drops, and the calculations are compared to measured volumes. For reliable volume predictions, the discrepancies regarding the input functions, $\theta(\phi)$ and $\zeta(\phi)$, need to be resolved. The experimental results from Part I showed the variation of the contact angles within a drop to be best fit by a third-degree polynomial taking the form

$$\cos \theta = 2 \frac{\cos \theta_{\max} - \cos \theta_{\min}}{\pi^3} \phi^3 - 3 \frac{\cos \theta_{\max} - \cos \theta_{\min}}{\pi^2} \phi^2 + \cos \theta_{\max}, \quad (41)$$

where $\theta_{\max}$ and $\theta_{\min}$ are the maximum and minimum contact angles of the drop, respectively. The base contour was found to take the shape of an ellipse, and a relation between the aspect ratio ($\beta = L/w$) and the Bond number was obtained empirically. With both the contact angle and the contour functions determined, the proposed method can be used to predict the volumes of drops on plane surfaces.

Verifying the method against experiments reported in the literature requires knowledge of dimensions that are rarely stated. However, Extrand and Gent [13] provided a top and a side image of a 3-mm³ drop with scales. The dimensions of the contour and the maximum and minimum contact angles were measured from the two images and used to define the $\theta(\phi)$ and $\zeta(\phi)$ functions as mentioned above. The drop was then reconstructed using the two-circle approximation. Fig. 7 shows the regenerated drop of Extrand and

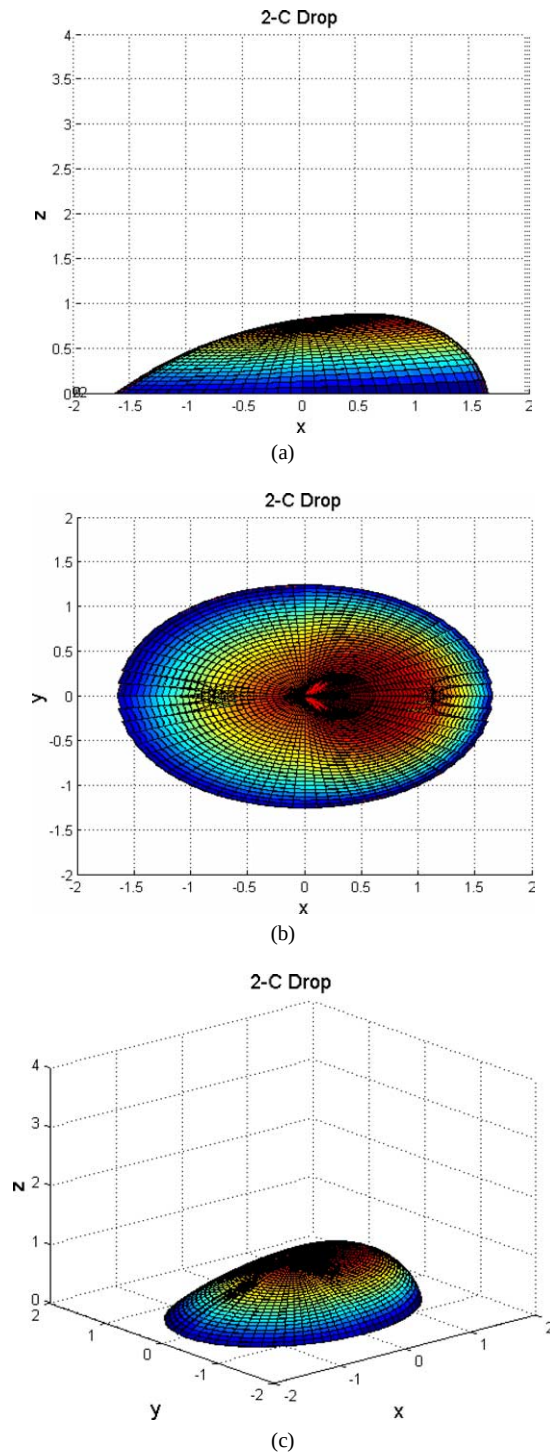


Fig. 7. A drop regenerated using the two-circle method. (a) Profile at the $\phi = 0$ plane; (b) contact line; and (c) three-dimensional view.

Gent. A three-dimensional model of the drop was also generated using stereolithography. The volume of the drop was predicted by the two-circle approximation with an error of 1%. The uncertainty in volume measurement was not stated in [13], but it is expected to be greater than 1%. Thus, the proposed method was successful in predicting the volume of the drop shown in [13].

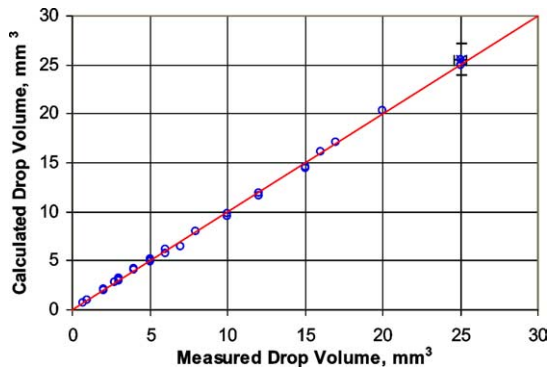


Fig. 8. Predicted drop volumes using the two-circle method versus measured values. The volumes were calculated from the dimensions of the elliptical contour and the maximum and minimum angles, taking $\cos(\theta)$ as a third-degree polynomial. $r^2 = 0.999$.

Experiments were conducted to test the validity of the proposed two-circle method under different conditions. A setup, described in Part I [11], was developed to allow recording contour and profile images of drops at different azimuthal and surface-inclination angles. In a typical test, a drop is injected through a microsyringe onto the test surface. A frontal image is recorded to determine the dimensions of the elliptical contour. Also, a side image is analyzed to obtain the maximum and minimum contact angles of the drop. Adopting the contact-angle function of Eq. (41), the volume of the drop was numerically computed using the two-circle method and compared to the volume measured in the microsyringe. Drops of water and ethylene glycol were tested on eight surfaces, providing a range of advancing contact angles from 49° to 112° , with receding angles of 12° to 78° . The tested drops ranged in volume from 0.7 to 25.0 mm^3 , with equivalent diameters from 1.18 to 5.80 mm. The maximum uncertainty in drop diameter was 2.3%. Surface inclination angles were varied from 0° (horizontal) to 90° (vertical). Fig. 8 shows the calculated volumes of 43 test drops, compared to the measured values. The method predicts the volumes of drops accurately with a coefficient of determination, r^2 , of 0.999. The small deviations are within the uncertainties associated with measurements of contour dimensions, contact angles, and volume. The close agreement between measured volumes and those predicted by the two-circle approximation validates the proposed method.

5.2. Sensitivity analysis

The two-circle method has been shown to provide reliable predictions of the volumes of liquid drops on plane surfaces. The proposed method can be used to study the effects of various approximations of drop shapes and parameters on volume prediction. An approximation of the drop shape will be discussed first, followed by the contour shape and the minimum contact angles.

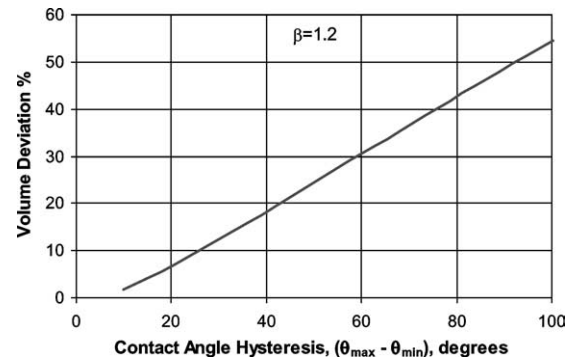


Fig. 9. Deviation of drop volume predicted by the spherical-cap approximation from the two-circle approximation.

5.2.1. Comparison to spherical cap approximation

The drop-shape problem is sometimes simplified by approximating the drop as a spherical cap. In this case, the contact angle is assumed constant around the circumference of the drop, and the contour is considered circular. Using this simplification, the drop volume is given by

$$V = \frac{\pi D^3}{24} \left(\frac{2 - 3 \cos \theta + \cos^3 \theta}{\sin^3 \theta} \right), \quad (42)$$

where θ is the average of $\theta_{\max}$ and $\theta_{\min}$. The spherical-cap approximation accurately predicts the volume of a drop on a horizontal surface, when the contour is circular and a single contact angle is observed, $\theta = \theta_{\max} = \theta_{\min}$. For such conditions, the two circles of the new method coincide into one, resulting in the same volume prediction. Thus, both methods yield accurate results for the special case of drops on horizontal surfaces. However, as a surface is inclined, the difference between $\theta_{\max}$ and $\theta_{\min}$ increases, and the use of an average contact angle causes larger errors in volume prediction. Therefore, the spherical-cap approximation becomes less accurate at higher contact-angle hysteresis. Fig. 9 presents the deviation of the spherical-cap volume of a drop from the two-circle volume, as a function of contact-angle hysteresis, $(\theta_{\max} - \theta_{\min})$. The calculations indicate that the spherical-cap approximation results in a 25% deviation in volume prediction when the contact angle hysteresis reaches 50° . Thus, the two-circle method provides a better approximation for the volume of a drop on a nonhorizontal surface.

The above comparison was made for identical drop parameters. ElSherbini [12] used both methods in further analysis to study drop parameters under critical conditions. The two approximations led to different contact angles and maximum diameters, which resulted in more pronounced deviations in predicting maximum volume. Although the spherical-cap approximation allowed deriving closed-form relations between critical parameters, the two-circle method provided more accurate predictions. Fig. 10 compares predictions of maximum drop volume, $V_{\max}$, using the two methods to measurements reported by Goodwin et al. [14]. The predictions are for water drops on vertical surfaces with given advancing contact angles. The figure shows that the

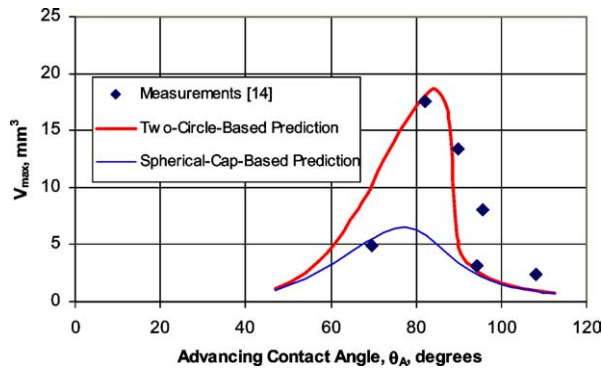


Fig. 10. Predictions of maximum volume of water drops on vertical surfaces compared to measurements by Goodwin et al. [14].

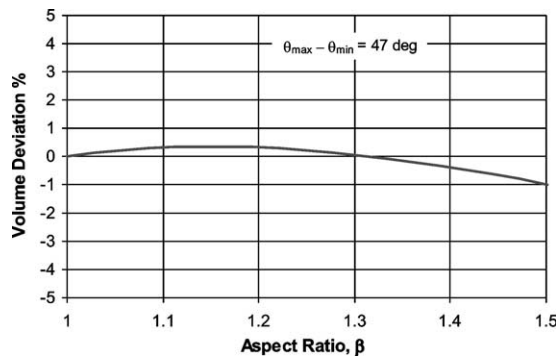


Fig. 11. Deviation in predicting drop volume when the elliptical contact line is approximated by a circular contour of the same area. This approximation results in an insignificant change in predicted volume.

analysis based on the two-circle method provides good predictions of $V_{\max}$. The spherical-cap approximation, however, can underestimate the maximum volume by up to 75%.

5.2.2. Effect of contour aspect ratio on volume

Experiments described in Part I show the general shape of a drop contour to be an ellipse.

The aspect ratio of the ellipse, which is the ratio of the major to the minor axis, depends on the Bond number, Bo . At $Bo = 0$, the aspect ratio, β , equals unity and the contour is circular. The aspect ratio increases with Bo to a maximum before the drop slides on the surface. The drop contour can be approximated by a circle having the same area as the ellipse. The error in drop volume resulting from this approximation is presented in Fig. 11 as a function of the aspect ratio. The deviation in predicted volume does not exceed 1%, confirming that the drop volume can be obtained with good accuracy when the contour is approximated by an equivalent circle. It is worth noting that this approximation leads to slightly larger errors in surface-tension forces acting on the drop, as discussed in [12].

5.2.3. Effect of minimum contact angle on volume

The experiments in Part I show that for a liquid drop on a nonhorizontal surface, the minimum contact angle, $\theta_{\min}$, changes between the characteristic angles θ_A and θ_R , de-

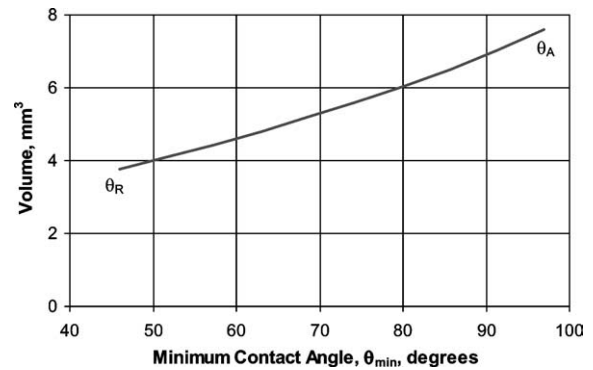


Fig. 12. Effect of varying the minimum contact angle of a drop from θ_R to θ_A on volume, keeping D and $\theta_{\max}$ constant. The assumption that $\theta_{\min} = \theta_R$ can change the volume by 50%.

pending on drop size and surface-inclination angle. In some investigations involving drops—such as studying heat transfer through wet surfaces or the mass of condensate retained on them—several researchers approximated the minimum angle, $\theta_{\min}$, as the receding contact angle, θ_R , to reduce the complexity of their models. The validity of such an assumption is checked using the two-circle method. The volume of a drop of given diameter and maximum contact angle is calculated for a range of $\theta_{\min}$ values, as presented in Fig. 12. The figure shows that substituting different values for $\theta_{\min}$ can significantly change the predicted volume. The assumption of $\theta_{\min} = \theta_R$ underestimates volume by as much as 50%. However, it remains a reasonable approximation for drops that are close to the critical size, since the assumption is true at incipient motion.

5.3. Using the proposed method to measure contact angles

The two-circle approximation of drop profiles can be used to analyze images of drops and quickly obtain reliable contact-angle measurements. Accurate measurements of contact-angles from drop images usually involve expensive software procedures to determine the slope of a curve fit to the profile. Instead, the contact angles can be found using the two-circle approximation, given the base length of the section, 2ζ , and the diameters of the two circles. Let d_1 and d_2 designate the diameters of the smaller and larger circles, respectively, in Fig. 3. Equations (21a) and (21b) can be rearranged to be

$$\sin \theta_1 = \frac{L_f}{1 + L_f} \frac{4\zeta}{d_1}, \quad (43a)$$

$$\sin \theta_2 = \frac{1}{1 + L_f} \frac{4\zeta}{d_2}, \quad (43b)$$

where, as in Eq. (20),

$$L_f = \frac{\sin \theta_1 (1 - \cos \theta_2)}{\sin \theta_2 (1 - \cos \theta_1)}. \quad (43c)$$

Equations (43) can be solved by simple iteration (or the Newton–Raphson method) to obtain θ_1 , θ_2 , and L_f . Notice that these equations are written in dimensionless form.

Therefore, the dimensions of ζ , d_1 , and d_2 can be used directly from the image without needing to scale them back to actual dimensions.

As an example, consider a drop image similar to that in Fig. 3. Notice that for the purpose of determining contact angles, the circles need to strictly fit the profile at the edges, but not necessarily at the middle of the drop. Let the drop base, 2ζ , measured from the image be 66 mm, with circle diameters $d_1 = 63$ mm and $d_2 = 92$ mm. Solution of Eq. (43) leads to $\theta_1 = 66.35^\circ$ and $\theta_2 = 53.85^\circ$. Thus, the two-circle approximation provides a convenient method for determining contact angles from side images of drops.

6. Conclusions

A method has been developed to approximate the three-dimensional shapes of liquid drops on vertical and inclined surfaces. The new method approximates a drop profile by parts of two circles that share a common tangent at the maximum height of the profile. The proposed approximation is based on experimental observations from Part I, although a physical explanation has been provided for the two-dimensional case of a drop on a vertical surface. For a given profile, the two circles can be completely defined in terms of the base length and the contact angles. The volume of a drop is then found by integrating the profiles over the circumference of the base. This approach leads to describing the volume of a drop, V , as a function of the contact angles and the base contour. Thus, the inputs to the two-circle method are the contact angle, θ , and contour radius, ζ , as functions of the azimuthal angle, ϕ . These functions are obtained experimentally in [11] to resolve disagreements among earlier works. When applied to images of a drop from the literature, the new method predicted its volume accurately. A physical model of the drop was then generated based on the proposed approximation of the shape. Measurements of drop volumes under various conditions showed that the two-circle method predicts volumes to within experimental uncertainties. The new method was used to study the sensitivity of

drop volume to some parameters and assumptions. Approximating the drop shape as a spherical cap can lead to a 25% error in volume at a high contact-angle hysteresis. Moreover, analysis of critical drops based on the spherical-cap approximation results in up to a 75% error in maximum drop volume. The effect of the contour aspect ratio on the drop volume was found to be negligible. The minimum contact angle, however, significantly influenced the accuracy of volume prediction. The two-circle approximation can be used to reliably measure contact angles from images of drop profiles.

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